



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year: 2023- 2024(OddSemester)

Active Learning Method

UNIT I – CLASSIFICATION OF SYSTEMS

Degree, Semester & Branch: B.E., III Semester & ECE 'A'

Course Code & Title: EC3354 & Signals and systems

Name of the Faculty member: Dr.S. Vairaprakash

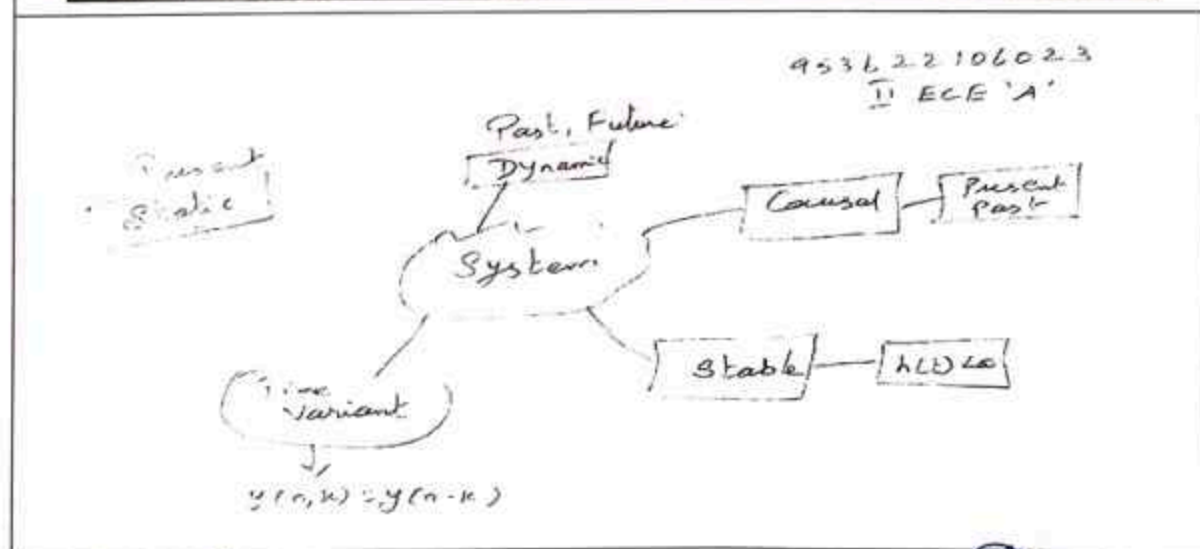
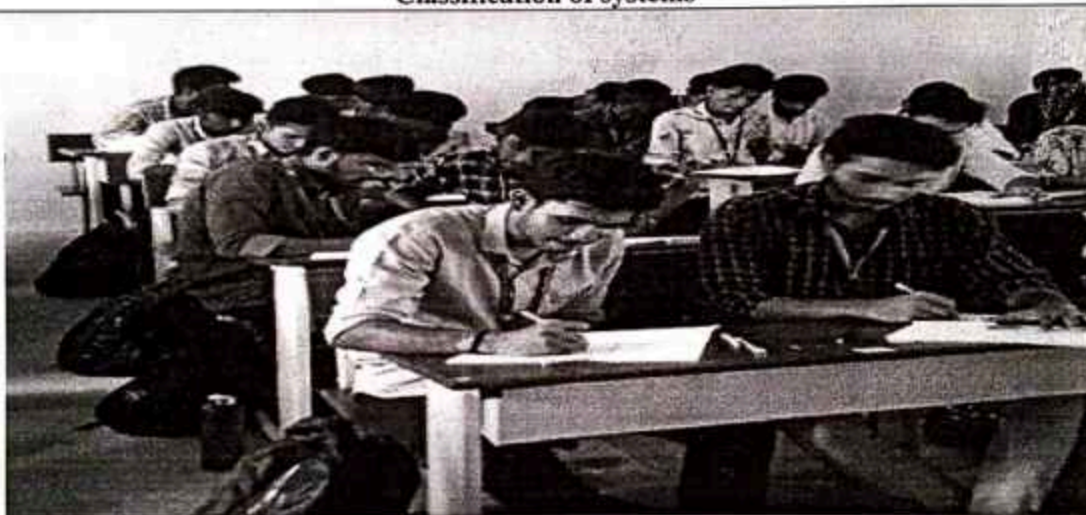
Name of the Topic: Classification of systems

Name of the Innovative Practice: Mind map

Date & Duration: 5.10.2023

Active Learning Method Execution

Classification of systems



[Signature] 6/10/23
Signature of the Faculty

[Signature]
HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year: 2023- 2024 (Odd Semester)

Innovative Teaching Method

UNIT III -LTI-Continuous Time system

Degree, Semester & Branch: B.E., III Semester & ECE 'A'

Course Code & Title: EC3354 & Signals and systems

Name of the Faculty member: Dr.S.Vairaprakash

Name of the Topic: Convolution Integral

Name of the Innovative Practice: One-minute Paper

Date & Duration: 28.10.2023 & 10 minutes

Sample Images:

Active Learning Method Execution

Convolution Integral



953622106301

ASWIN VINEX

III SEM ECE 'A'

Convolutional Integral.

$$* \int x(\tau) h(t-\tau) d\tau =$$

$$* X(s) H(s)$$

$$* H(s) = \frac{Y(s)}{X(s)}$$

Signature of the Faculty

HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year: 2023- 2024(Odd Semester)

Active Learning Method

UNIT IV – Analysis of Discrete Time signals

Degree, Semester & Branch: B.E., III Semester & ECE 'A'

Course Code & Title: EC3354 & Signals and systems

Name of the Faculty member: Dr.S. Vairaprakash

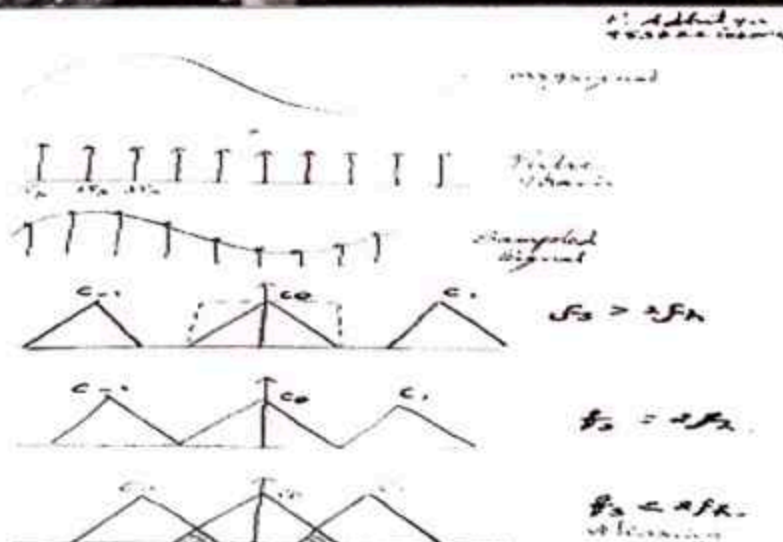
Name of the Topic: Signal Sampling and Aliasing

Name of the Innovative Practice: Sketch Noting

Date & Duration: 22.11.2023 & 10 minutes

Active Learning Method Execution

Signal Sampling and Aliasing



[Signature]
23/11/23
Signature of the Faculty

[Signature]
HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.E., III Semester & ECE A

Course Code & Title: EC3354 & Signals and Systems

Name of the Faculty member (s): Dr.S.Vairaprakash

Innovative Practice Description

- **Unit / Topic:** II/Laplace Transform Properties
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO7
- **Activity Chosen:** Flipped Class room
- **Justification:**

The topic was chosen for flipped class because Laplace Transform has eight properties.. The Unit 2 deals Fourier Transform and Laplace Transform applications to signals in Frequency domain using various properties. Since the students are exposed to the Laplace transform in mathematics and circuit analysis ,the topic of properties in Laplace transform for flipped class activity was given . This topic can be learnt in self-learning mode with simple derivations and logic. The students have chance to browse the topic various source and through mathematics concept. And find more information about how to make use the properties in problem solving instead of using formula .. Thus this flipped class activity is chosen to make them to learn easy by work out different properties .

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The overview of flipped class activity was given to students before the implementation of the activity. The students were divided into groups and one property is allotted to each group from Laplace transform topic as their wish. The topic related information and e-sources were given to the students. The students have to prepare a presentation for 10 minutes and each group has to present their topic chosen and others will listen to the presentation. After the presentation, question & answering is planned and others can share their ideas and doubts. By this way, all groups were shared their concepts learnt in their topic inside the class .

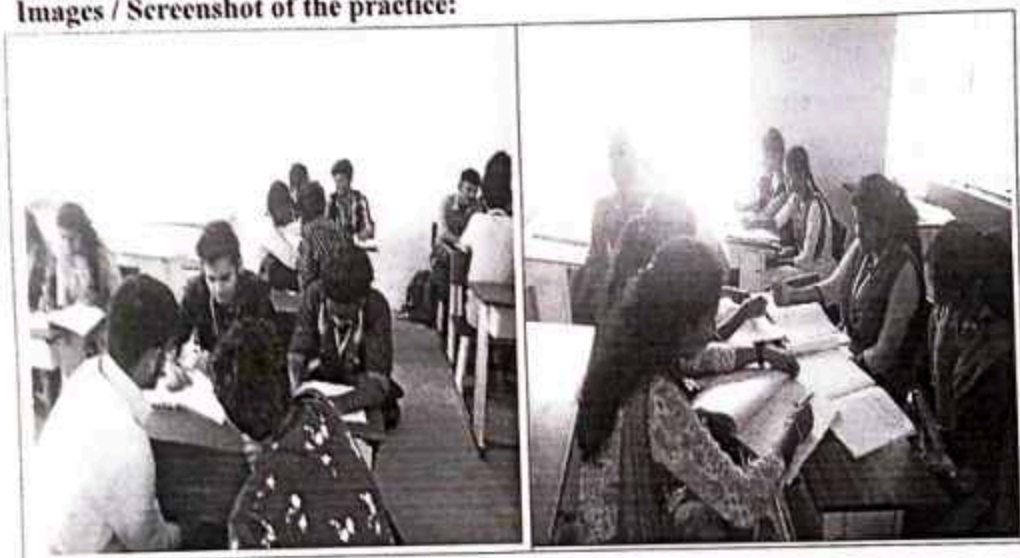
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO8	PO9	PO10	PSO1	PSO3
CO3	3	2	2	2	3	2	3	2

• PO / PSO mapped:

	PO9	PO10	PSO3
Innovative practice	3	2	3
Justification for correlation	The students work as a member of a team to achieve the goal. So, the course outcome is highly correlated.	Through flipped classroom activity the students' listen to, comprehend information, instructions and viewpoints of others.	The students will be able to innovate new ideas to solve existing/novel problems in the analysis of CT signals.

• Images / Screenshot of the practice:



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The students felt a new practice and it helped them to know to make good presentation and avoid stage fear. They liked this practice of teaching that can be done through this type of activity. They could get idea from learning sources which would help them to prepare the presentation.

❖ *Benefit of the practice:*

- ❖ Students completed this task successfully. They conceptually prepared the presentation and organized the exercise for the topic they had selected. Every member of the group had a good understanding of the subjects. Students were able to understand concepts by discussing thoughts within the group they formed. Every group member has the ability to explain and convey the idea of properties.

❖ *Challenges faced in implementation:*

Some of the students felt difficult to present because of lack of communication.

Few groups done their presentation beyond their time limit.

Few groups faced the issue in presentation because of lack of preparation

Few students in the group felt difficult to answer all the questions.

References:

Textbooks:

1. Allan V. Oppenheim, S. Wilsky and S.H. Nawab, "Signals and Systems", Pearson Education, 2007.

Other References:

1. B. P. Lathi, "Principles of Linear Systems and Signals", Second Edition, Oxford, 2009.
2. R.E. Zeimer, W.H. Tranter and R.D. Fannin, "Signals & Systems - Continuous and Discrete", Pearson, 2007.
3. John Alan Stuller, "An Introduction to Signals and Systems", Thomson, 2007
4. P. Ramesh Babu and R. Anandanatarajan, "Signals and Systems", SciTech publications, Fourth Edition.
5. A. Nagoor Kani, "Signals and Systems", Tata McGraw Hill Education Private Limited, Second Edition, 2010.
6. <https://ctl.utexas.edu/instructional-strategies/flipped-classroom>


Signature of the Faculty


HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.E., III Semester & ECE A

Course Code & Title: EC3354 & Signals and Systems

Name of the Faculty member (s): Dr.S.Vairaprakash

Innovative Practice Description

- **Unit / Topic:** III/ Differential Equation- Laplace transforms in Analysis of CT systems
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Round Table Activity
- **Justification:**

A round table activity is a collaborative learning approach where participants gather in a circular arrangement to discuss, share ideas, and collectively address a specific topic or problem. This format is designed to encourage equal participation, open communication, and the sharing of diverse perspectives. Round table activities are commonly used in educational settings, business meetings, and community discussions to foster engagement and collaboration.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

Introduction (Facilitator):

Begin by introducing the purpose of the activity and its relevance to CT systems analysis. Briefly review Laplace transforms and their significance in solving differential equations.

Present the Problem (Facilitator):

Share the selected differential equation representing the CT system.
Discuss the components of the equation and any relevant parameters.

Round 1 - Understanding the Problem (Participants):

Allow participants to express their understanding of the problem.
Encourage questions and clarifications to ensure everyone is on the same page.

Round 2 - Laplace Transform Application (Participants):

Guide participants through the application of Laplace transforms to the given differential equation.

Discuss the transformation of each term and how it simplifies the equation.

Round 3 - Solving in Frequency Domain (Participants):

Lead a discussion on solving the resulting algebraic equation in the frequency domain.
Emphasize the steps involved in isolating the Laplace-transformed output variable.

Round 4 - Inverse Laplace Transform (Participants):

Discuss methods for finding the inverse Laplace transform to obtain the time-domain solution.
Explore Laplace transform tables and techniques for inverse transforms.

Round 5 - System Analysis (Participants):

Analyze the time-domain solution and discuss its implications for understanding the CT system's behavior.

Explore the effects of different input functions or initial conditions.

Summary and Closing (Facilitator):

Summarize key insights and learning points from the activity.

Encourage participants to ask final questions or share reflections.

Feedback: Collected feedback from participants to improve future round table activities.

CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO8	PO9	PO10	PSO1	PSO3
CO3	3	2	2	2	3	2	3	2

PO / PSO mapped:

Innovative practice	PO9	PSO1
	3	3
Justification for correlation	The course outcome is highly correlated since the students work well as individuals and as a team to solve differential equation representing the CT system while doing the round table activity..	The students come up with new ways to solve existing/new problems while determining the response to an LTI CT system.

- Images / Screenshot of the practice:**



- Reflective Critique:**

- ❖ *Feedback of practice from students and other stakeholders:*

The students felt that their group worked well together, and everyone had an opportunity to express their thoughts. The round table format created a collaborative environment, and we were able to learn from each other. Addressing challenges within the group enhanced problem-solving skills. The problem statement was presented clearly, and the facilitator's guidance was helpful. Working in a group helped me understand different perspectives on solving the problem.

❖ ***Benefit of the practice:***

By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. The round table activity offers a multifaceted approach to learning, promoting engagement, collaboration, and critical thinking while creating an environment that prepares students for future challenges.

❖ **Challenges faced in implementation:**

Some students may not adequately prepare for the round table activity, leading to a less effective discussion.

Balancing the time allocated for the activity with the need to cover the required content can be challenging.

Some students may resist collaborative learning and prefer individual study.

References:

Textbooks:

1. Allan V. Oppenheim, S. Willsky and S.H. Nawab, "Signals and Systems", Pearson Education, 2007.

Other References:

1. B. P. Lathi, "Principles of Linear Systems and Signals", Second Edition, Oxford, 2009.

2. R.E. Zeimer, W.H. Tranter and R.D. Fannin, "Signals & Systems - Continuous and Discrete", Pearson, 2007.

3. John Alan Stuller, "An Introduction to Signals and Systems", Thomson, 2007

4. P. Ramesh Babu and R. Anandanatarajan, "Signals and Systems", SciTech publications, Fourth Edition.

5. A. Nagoor Kani, "Signals and Systems", Tata McGraw Hill Education Private Limited, Second Edition, 2010.

6. <https://educators.brainpop.com/teaching-tip/roundtable-learning-strategy/>


Signature of the Faculty


HOD